Lost in Conversation

Comparing Memory Strategies in Multi-Turn Conversations

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The Problem : Remembering User Requirements

- Users share information throughout a conversation.
- They may change their preferences or correct earlier details.
- In long conversations, important information can be missed or incorrectly updated.
- A dialogue system needs to remember the latest requirements.

Research Question and Approaches

How do different memory strategies affect the retention and correction of user requirements in multi-turn dialogues?

Comparing three dialogue memory strategies:

Full History

Summary Memory

SISU

Dialogue Datasets

Set 1	19 turns	Synthetic	Simple scenario
Set 2	50 turns	Synthetic	Multiple corrections
Set 3	79 turns	Human-AI Conv.	Hesitations, interruptions, corrections

Methods

Full History

• Store all dialogue turns in conversation history.

• Send the complete history to Llama 3.1 to extract the final requirements.

Rolling Summary

• Update a rolling summary after every two dialogue turns.

• Send the final summary to Llama 3.1 to extract requirements.

SISU

• Interpret dialogue turns as moves and update an Information State.

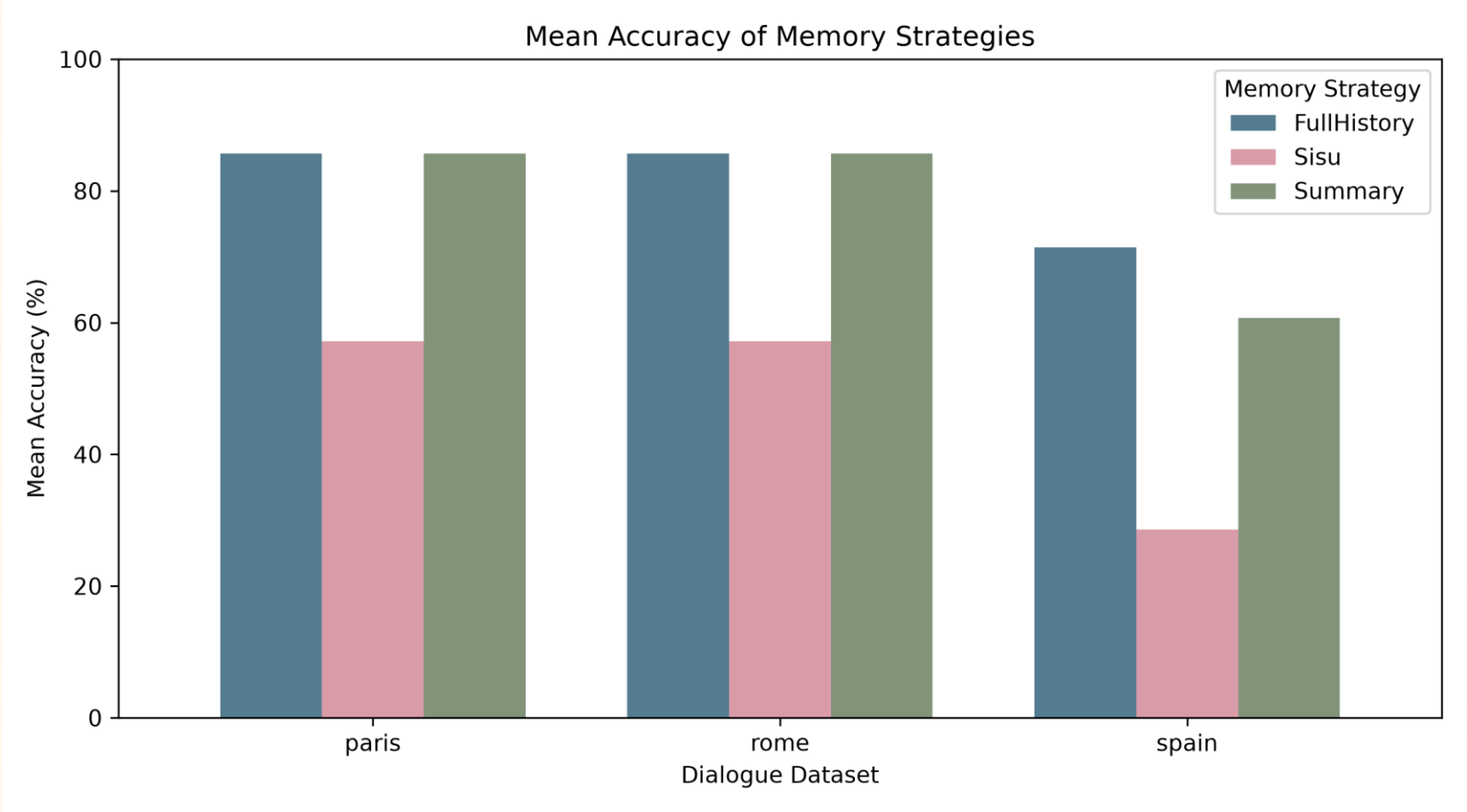
• Store requirements in commitments (com) and track questions using QUD.

Evaluation

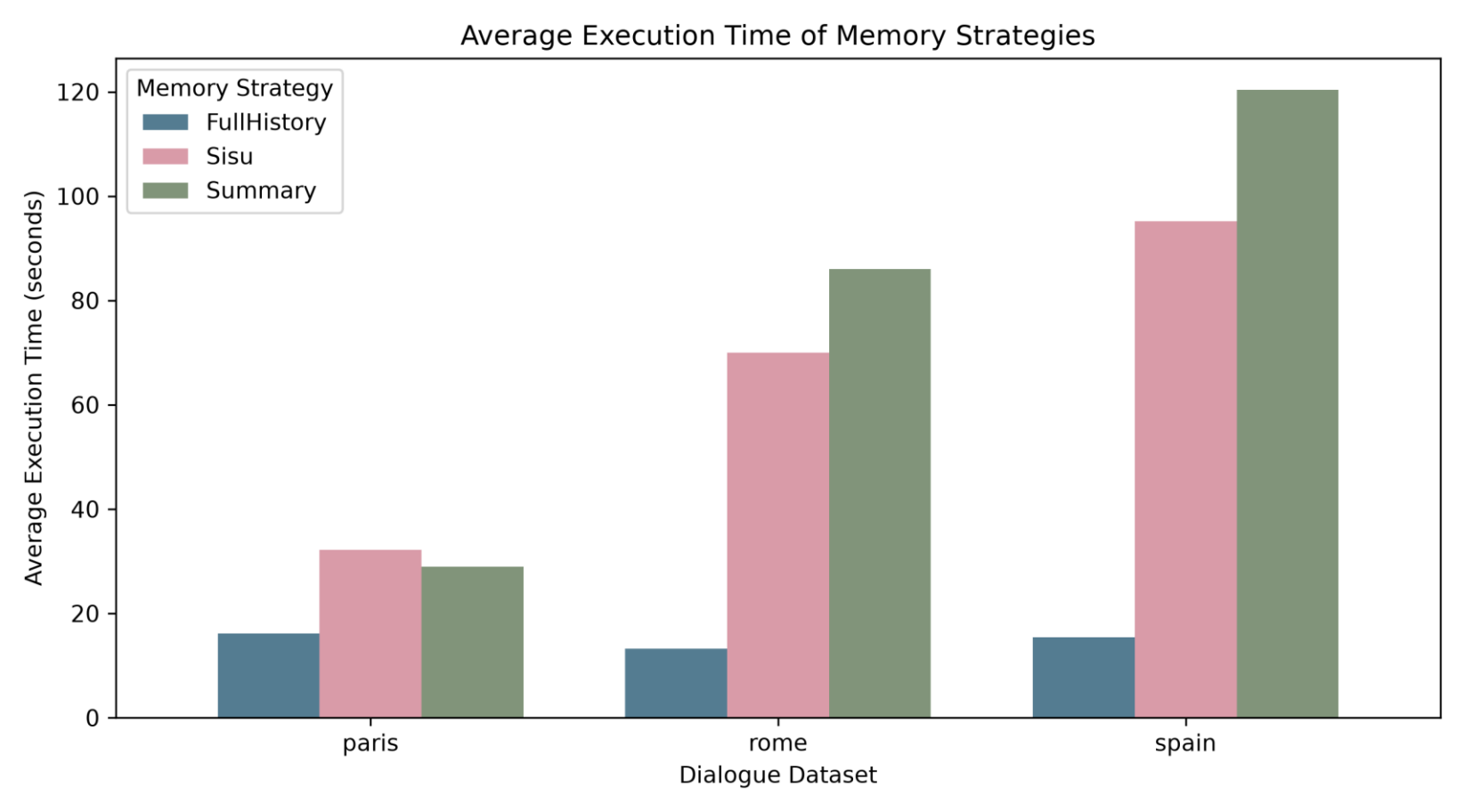
- Extract the final seven travel requirements.
- Compare predicted requirements with manually defined ground truth.
- Use strict matching to calculate accuracy.
- Run each strategy five times on each dataset.

Accuracy = (Correct requirements / Total requirements) × 100

Accuracy Comparison Across Memory Strategies



How Long Does Each Strategy Take?



Key Findings and Limitations

- Full History performed best overall.
- Summary Memory sometimes lost or added information.
- SISU struggled with interpreting and updating information.
- Small dataset: only three dialogues.
- Strict matching misses similar meanings.
- Different numbers of LLM calls affect runtime comparisons.

Future work

- Expand the datasets
- Improve evaluation
- Integrate with speech state

Reference

LLMs Get Lost In Multi-Turn Conversation

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Thank you